

1. Administrative & Study Identification

Full Study Title: A Multicenter, Randomized, Double-Blind Placebo-Controlled Phase 3 Study to Evaluate the Pharmacokinetics, Efficacy and Safety of Deucravacitinib (BMS-986165) in Pediatric Subjects With Moderate to Severe Plaque Psoriasis **Short Title/Acronym:** PK, Efficacy and Safety of BMS-986165 in Pediatric Moderate to Severe Plaque Psoriasis **Protocol Number:** IM011-126 **NCT Number:** NCT04772079 **Sponsor Name:** Bristol-Myers Squibb **Investigational Product:** Deucravacitinib (BMS-986165) / Placebo **Phase of Development:** Phase 3 **Medical Monitor:** Head of Global Submission Management - Clinical Trials (Bristol-Myers Squibb)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the drug levels (pharmacokinetics), efficacy, and safety of Deucravacitinib in pediatric participants aged 4 to <18 years with moderate to severe plaque psoriasis. **Secondary Objectives:** To determine optimal body-weight-based dose levels via pharmacokinetic assessment (Part A), to establish the clinical efficacy and safety of standard and half-standard doses versus placebo (Part B), and to monitor long-term safety and tolerability over a 5-year long-term extension (LTE) period.

2.2 Background & Rationale To address the need for orally effective, targeted therapeutics with improved safety profiles for pediatric patients suffering from moderate to severe plaque psoriasis. Deucravacitinib, a selective Tyrosine Kinase 2 (TYK2) inhibitor, has demonstrated significant efficacy and safety in adults. This study aims to provide definitive clinical data and dosing guidelines to expand therapeutic options for the pediatric population, a group often limited by the toxicities or injection requirements of existing systemic therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (3 Arms in Part B: Standard dose, half-standard dose, and placebo)

3.2 Planned Enrollment & Duration Trial Status: Recruiting **Target** **\$N\$:** 153 participants **Number of Sites:** ~36-37 global locations

Estimated Study Start: 23-Mar-2021 **Estimated Primary Completion:** 01-Jan-2029 (Estimated Study Completion: 08-Sep-2033 for the LTE)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Males and females aged 12 to <18 years for Cohort 1; Males and females aged 4 to <12 years for Cohort 2. 2. Confirmed diagnosis of moderate to severe plaque psoriasis for at least 6 months. 3. Candidate for phototherapy or systemic therapy. (*Note: Must complete the Week 52 treatment period in Part A or B to enter the long-term extension [LTE] period*)

4.2 Exclusion Criteria 1. Participants weighing ≤ 30.0 kg at screening for Cohort 1 (age 12 to < 18 years); Participants weighing ≤ 18.0 kg at screening for Cohort 2 (age 4 to < 12 years). 2. Presence of other forms of psoriasis or a history of recent infection. 3. Prior exposure to deucravacitinib (BMS-986165) or active comparator; evidence of active Tuberculosis (TB) for LTE period.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Once-daily weight-based oral dosing. In Part A, patients receive 4mg or 6mg for 2 weeks to evaluate processing. In Part B, patients are randomized to receive a standard weight-based dose, a half-standard weight-based dose, or placebo. **Route of Administration:** Oral **Duration of Treatment:** Up to 52 weeks for the main study period, followed by an optional 5-year long-term extension (LTE).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for psoriasis that does not interact with the investigational product. **Prohibited:** Prior exposure to deucravacitinib or concurrent use of active comparators/other systemic psoriasis medications during the double-blind phase.

6. Study Timeline & Visit Schedule

Visit ID	Window	Key Activities/Procedures
Screening	Days Prior to Day 0	Informed Consent/ Assent, Medical History, Weight confirmation, Baseline severity evaluation.
Baseline (Part A/B)	Day 0	Randomization, First Dose, Pharmacokinetic (PK) Blood Sampling initialization.
Interim 1 (Part A)	Week 2	PK assessment for dose selection; transition into up to 52 weeks of continuous treatment.
Interim 2 (Part B)	Week 16 & Week 32	Efficacy assessment. At Week 16, placebo non-responders and partial-responders on lower doses are adjusted/re-randomized.
End of Study	Week 52 / 60	End of primary treatment, safety follow-ups at Week 52 and 60, followed by transition to 5-year LTE.

(Note: Part A patients have up to 19 treatment visits; Part B patients have up to 15 treatment visits)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1. Geometric mean observed average concentration at steady state (C_{avg.ss}) for deucravacitinib at Week 2: Part A. 2. Maximum observed plasma concentration at steady state (C_{max.ss}) for deucravacitinib at Week 2: Part A. 3. Trough observed plasma concentration (C_{trough}) for deucravacitinib at Week 2: Part A. (Also involves clinical efficacy and safety defined by the proportion of participants achieving targeted clearance (e.g., PASI 75 and sPGA 0/1) at Week 16 (Part B)). **Measurement Method:** Serum PK analysis and standardized clinical severity scoring (PASI and sPGA) by Investigators.

7.2 Secondary & Exploratory Endpoints 1. Incidence of Adverse Events (AEs). 2. Incidence of serious adverse events (SAEs). 3. Incidence of clinically significant changes in clinical laboratory results: Hematology tests. (Additionally includes evaluation of high-level clearance rates (PASI 90, PASI 100) and Quality of Life improvements (DLQI), and long-term safety, tolerability, and sustained response during the 5-year LTE).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Conducted continuously across the 15-19 clinical visits and during the 5-year LTE. Evaluates specific AEs of interest such as upper respiratory tract infections or nasopharyngitis. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeline to the sponsor and Institutional Review Boards. **Data Monitoring Committee (DMC):** Independent/Internal Safety Review Committee to assess Part A PK/safety data before determining the appropriate standard/half-standard doses to proceed into Part B.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy evaluation; Safety Set for all participants receiving at least one dose of the investigational product. **Sample Size Justification:** \$N = 153\$ adequately powered to evaluate pharmacokinetics in the pediatric cohorts and demonstrate statistically significant efficacy differences between the active dose arms and placebo. **Handling of Missing Data:** Standard non-responder imputation (NRI) for binary clinical endpoints (PASI/sPGA).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite clinical monitoring combined with centralized data review ensuring pediatric trial

safeguarding. **Audit Trail:** eCRF locking and query resolution handled via standard sponsor data clarification forms.

11. Study Identifiers & Governance

Primary ID: IM011-126 **Posting ID:** 4322140572307079 **Registry**

Number: NCT04772079 **Sponsor/Lead Organization:** Bristol-Myers

Squibb **Study Title:** A Study to Evaluate the Drug Levels, Efficacy and Safety of Deucravacitinib in Pediatric Participants With

Moderate to Severe Plaque Psoriasis **Official Regulatory Title:** A

Multicenter, Randomized, Double-Blind Placebo-Controlled Phase 3

Study to Evaluate the Pharmacokinetics, Efficacy and Safety of

Deucravacitinib (BMS-986165) in Pediatric Subjects With Moderate

to Severe Plaque Psoriasis

12. Clinical Framework (Psoriasis Specific)

Primary Condition: Moderate to Severe Plaque Psoriasis **Diagnosis**

Threshold: Diagnosis of plaque psoriasis for ≥ 6 months

Medical Methodology: Selective Tyrosine Kinase 2 (TYK2) Systemic

Inhibition **Optimization Target:** PASI 75 (75% improvement in

Psoriasis Area and Severity Index) and sPGA score of 0 or 1

(clear or almost clear)

13. Study Procedures & Methodology

Management Pathway: Oral targeted systemic therapy vs. Placebo

Education Protocol (HCP): Site initiation and training for pediatric PK sampling and assessment scoring (PASI/sPGA).

Education Protocol (Patient): Onsite training/caregiver instruction for daily oral medication adherence and weight-

monitoring tracking. **Quality Audit Mechanism:** HCP adherence to the weight-based dosing guidelines and clinical assessment

synchronization.

14. Observation & Follow-Up Schedule

Total Duration: Up to 60 weeks (Primary treatment + follow-up) followed by a 5-year LTE **Onsite Visit Cadence:** Up to 19 visits

for Part A; up to 15 visits for Part B **Remote Monitoring**

Frequency: As necessary between scheduled clinic visits.

